Course Project form

Course Code & Title:

COME413 COMPUTER NETWORKS

Course Project Proposal Title:

SOCKET PROGRAMMING IN PYTHON

Project Description: (maximum 350 words)

Computer networks form the essential backbone of modern software systems, relying on robust communication protocols to ensure seamless data exchange across distinct devices. The primary objective of this project is to develop a reliable, scalable, and interactive network communication application based on a client-server architecture using the Python programming language. This system will apply fundamental networking principles in a practical framework, demonstrating exactly how data is routed, error-checked, and safely delivered over a local area network or the internet.

Moving well beyond a superficial message-passing approach, this project will emphasize the strict implementation of communication protocols and system fault tolerance. The development phase will heavily utilize object-oriented programming (OOP) principles to construct a modular, readable, and highly maintainable codebase. Core algorithms will be optimized for efficient data serialization and packet parsing. To prevent issues inherent to data transmission, such as packet loss, transport layer standards will be implemented to ensure that data arrives accurately and in the correct sequence.

A critical component of managing network traffic effectively is the integration of multi-threading and asynchronous operations. By leveraging these techniques, the server architecture will seamlessly handle multiple concurrent client connections without blocking the main execution flow. Furthermore, unexpected network scenarios—such as sudden connection drops, socket timeouts, or invalid network requests—will be managed through comprehensive algorithmic exception handling, preventing application crashes.

Another integral part of this project is the development of a fully functional Graphical User Interface (GUI). Through this responsive GUI, users will be able to dynamically configure crucial network parameters, including target IP addresses and port numbers. The interface will allow users to monitor active connection statuses and visually track bidirectional data transfers in real-time.

Ultimately, this project aims to bridge theoretical computer networking concepts with modern software engineering practices, resulting in a robust, professional-grade application that thoroughly simulates real-world socket programming operations.

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